

CORVUS SENTINEL

DETECT. CLASSIFY. ALERT. | On-device acoustic drone detection

Old Crows Wireless Solutions (OCWS) · Sideloadable Android · Offline · No extra hardware required

WHAT IT IS

Corvus Sentinel turns a phone or tablet into a passive, listen-only early-warning sensor for drones. It detects rotor acoustic signatures through the device microphone, classifies them on-device (no internet, no data leaving the device), tracks contacts, and briefs the operator. It is designed for the operator who needs awareness now on gear they already carry — not a six-figure fixed radar installation.

CURRENT CAPABILITIES

Detect & classify	Acoustic classifier (None / Skydio X2 / DJI Phantom / Parrot Anafi / Unknown), on-device.
Homemade flag	Open-set logic flags a contact that is clearly a drone but matches no library profile as “ possible homemade / unknown build ” — the threat a database-only detector misses.
Range	Loudness-based range band (e.g. ~150–250 ft). Single mic = no bearing ; a manual rotate-to-peak aid gives coarse direction.
Tag & report	GPS-stamped, timestamped intercepts and an exportable After-Action Report.
Background	Keeps monitoring when minimized; heads-up notification on every new contact.
Remote ID (RF)	Receives drone Remote ID over Bluetooth — drone position and operator/pilot location — for compliant drones.
Tactical map	Offline-capable map: operator, Remote ID drone/pilot pins, and acoustic range rings.
Learning library	Captures unknown/homemade contacts for labeling and retraining — the system improves from field encounters.

HONEST LIMITATIONS (read before testing)

- The bundled model is trained on **physically-grounded synthetic** signatures. Real-world accuracy is **not yet validated** — that is exactly what this testing is for.
- Acoustic detection wins on cost/accessibility, not range/precision; it does not replace radar/RF.
- **No bearing or elevation** from a single mic; range is a coarse estimate.
- Remote ID only sees **compliant** drones that broadcast it; homemade/hostile drones will not appear there (caught acoustically).
- External RF (LoRa / ExpressLRS / control-link) detection needs an add-on SDR module — scaffolded but disabled until hardware.

WHAT WE'RE ASKING YOU TO TEST

- Install the APK (sideload, no account) and confirm it runs: mic permission, level meter, START.
- Fly known drones at marked distances — does it fire correctly, and does “None” hold in quiet?
- Note detection rate, false positives, and effective range — help us build a real confusion matrix.
- Try the **homemade flag** with a non-standard / FPV build if available.
- With a recent DJI, check **Remote ID** shows drone + operator location.
- Verify **background** detection + notifications with the screen off.
- Capture clips of anything interesting — they feed retraining.

ROADMAP

External RF module (SDR) for LoRa/ExpressLRS & control-link detection · stereo/array direction-finding · real-data retrained model · ATAK / Cursor-on-Target output · rugged tactical-phone (ATAC-class) integration.

Corvus · Old Crows Wireless Solutions · We Always Find the Signal. | Contact: Joshua · joshua@oldcrowswireless.com | Pre-release engineering build — for authorized evaluation only.